Nader Hegazy

Senior Software Engineer (Tools & Pipeline)

eng.nadernour@gmail.com | Portfolio | LinkedIn | GitHub | YouTube

Professional Summary

Senior Software Engineer with **11+ years of experience** specializing in **C++, Tools Architecture, and Pipeline Automation**. Proven track record of accelerating production workflows and eliminating technical bottlenecks for industry leaders like **Epic Games** (Twinmotion) and **Adidas**. Expert in architecting high-performance C++ plugins, scalable content pipelines, and intuitive UI/UX for internal tools. Dedicated to empowering technical artists, reducing iteration times from hours to seconds and unblocking global production pipelines.

Professional Experience

Epic Games (via 4D Pipeline) | Senior Software Engineer (Twinmotion)

Jan 2023 – Mar 2026 (3.25 Years)

- **Real-Time Telemetry:** Architected a Statistics Monitor (Overview, Textures, Meshes) processing telemetry for heavy scenes, empowering users to identify bottlenecks and replace assets on-the-fly.
- **UI Performance Optimization:** Migrated Material Panel WrapBox to TileView (C++, Slate/UMG), reducing load times for heavy architectural scenes (5,000+ materials) from 15 minutes to ~0.5 seconds while preserving complex interactions.
- **Engine Migration & Profiling:** Resolved critical engine regressions across 5 major Unreal Engine version upgrades (UE 5.1 to 5.6). Utilized Unreal Insight and RenderDoc to identify regressions, rescuing unplayable scenes by restoring framerates from ~2fps to a workable ~30fps.
- **Apple Silicon Native Support:** Engineered the Mac Arm64 packaging pipeline by modifying underlying Twinmotion build scripts, delivering native M1/M2 support and a 20-30% performance increase over Rosetta.
- **Feature Engineering:** Engineered high-end Cinematic Camera features (adjustable lag, Focus Hunting) to unblock artist workflows, and built a robust Undo/Redo system for the material system that prevented data loss.
- **Automated Testing:** Built comprehensive C++ testing suites for scene serialization and Undo/Redo state management, eliminating manual QA overhead and catching regressions pre-deployment.

Adidas (via 4D Pipeline) | Senior Software Engineer (Main Developer)

Jun 2021 – Dec 2022 (1.5 Years)

- **Adidas CLO Plugin:** Consolidated legacy tools into a unified, high-performance asset library (C++, Qt, CLO-SDK), enabling hundreds of global designers to seamlessly drag-and-drop assets within their 3D environment.
- **Pipeline Automation:** Engineered a custom Unreal Engine Alembic Importer Plugin, reducing asset import time from ~15 minutes to a single click, dramatically accelerating throughput.
- **Cloud Architecture:** Architected and migrated the central 3D asset database to a scalable AWS Serverless backend (Python, Lambda, Step Functions, Event Bridge, S3), successfully supporting tens of thousands of digital garments across global "Digital Creation" hubs.
- **Data Synchronization:** Built a specialized Smartsheet API Tool for automated asset data synchronization using JSON-templated CSV generation, saving 99% of manual data entry time.

Instinct Games | Senior Software Engineer (Steel Rising)

Dec 2017 – May 2021 (3.5 Years)

- **Complex Tooling:** Architected a CSV Item Generation tool powering the procedural economy, enabling designers to dynamically generate and balance thousands of asset variations (robots, weapons, materials).
- **Pipeline Automation:** Engineered a C++ tool to resolve mesh clipping across hundreds of multi-layered outfits by procedurally calculating vertex offsets, eliminating hundreds of hours of manual artist authoring.
- **Networked Physics:** Engineered a physics-driven, rideable hoverbike, optimizing UE4 network replication to ensure smooth, synchronized physics across multiplayer clients.
- **Engine Architecture:** Performed deep UE4 source code modifications (C++) to resolve critical bugs and led the migration of a highly customized engine build to newer releases, successfully unlocking rendering features for the production team.

ADabisc Future Qatar | Software Engineer

Dec 2016 – Jul 2017 (8 Months)

- Developed a real-time face tracking prototype using an Intel RealSense camera and dlib to map facial markers to 3D character blend shapes.
- Engineered a Unity-based drawing application utilizing custom C++ native plugins for performance-critical fill algorithms.

Al-Kottab | Game Developer

Jan 2015 – Dec 2016 (2 Years)

- Rapidly prototyped and delivered AR/VR experiences, including a VR multiplayer racing game with custom driving AI and collision avoidance.
- Successfully advocated for and led the studio's transition to Unreal Engine 4 for high-fidelity VR development.

Technical Skills

- **Languages:** C++, Python, C#, SQL, Kotlin, JavaScript, Java, Lua.
- **Engines & Frameworks:** Unreal Engine 4/5 (Core, Rendering, Slate/UMG), Unity, Qt, Jetpack Compose, Clo-SDK, .NET, Vulkan, OpenGL.
- **Tools & DevOps:** Perforce, Git, SVN, CMake, Premake, Jenkins, Jira, RenderDoc, Unreal Insight, Linux, Custom Toolchains.
- **Specializations:** Performance Optimization, Plugin Architecture, Cross-Platform 3D Architecture, UI/UX for Tools, Computer Graphics (PBR, Shaders), VR/AR (Vuforia, Oculus Rift).
- **Cloud & Backend:** AWS (Lambda, Step Functions, Event Bridge, S3), Node.js.

Education

B.S. in Computer Engineering, Ain Shams University, Egypt.